



CSE251: Electronic Devices and Circuits

Lecture 0

Introduction to Electronic Devices

Prepared By:

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Outline

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Instructor information

- Shadman Shahid [HAD]
- Seat No. – 4N159:
 - Consultation: **SUNDAY – 11 AM to 2 PM**
- Reachable via mail or discord:
 - Mail: ext.shadman.shahid@bracu.ac.bd
 - Discord: shadman<dot>shahid
- Research Interest:
 - Photonics in computational devices

Administrative Details

- Course Discord: Will be updated soon.
- Course Drive folder: [CSE251-Spring24-HAD](#)
 - Course Handout – Syllabus, Grading Policy.
 - Course Calendar
 - Homework Assignments
 - [Past Exams](#) and [Practice Problems](#)
 - Class Notes
 - [Recorded Lectures](#)

Marks Distribution

Assessment	Percentage	Total number of assessments	Number of assessment to be graded
Attendance	8%	-	-
Assignment	12%	5	Best N-1
Quiz	15%*	4*	Best N-1
Midterm	20%	1	1
Final	20%	1	1
Lab	25%	-	-

*I will take pop quizzes every now and then at the **end** or **beginning** of a class to be added as **2%** bonus mark.

Percentage of Classes Attended	Marks
above 70	8
65-69	7.5
60-64	7
55-59	6.5
50-54	6
45-49	5
40-44	4
below 40	0

PS: Bonus will **only be added** to Assignment and Quiz marks. If you obtain the designated **27%** in quiz and assignments, the bonus will not be added to other areas.

How to get an A/A+ in CSE251?

Time Management: Allocate 10 hrs/wk of regularly scheduled times in the week outside of class for CSE251:

- 30 min for **reading** of textbook / slides **before** each class
- 30 min for **studying** online notes **before** each class
- 30 min for **studying** these notes **between** classes
- 75 min for **practicing** problems (Check the practice sheet and previous questions).
- 4-5 hrs/wk for HWs / Assignments.

– In a semester, all lectures total only 30 hrs, which is less than 1 week at a job! It's up to you to put in the time to learn

– Get a 1" binder (organize lecture notes/HWs/exams)

– Start assignments early. **Do all problems by yourself first.** If you get stuck, form study groups to work on problems together but **ALWAYS** write-up and submit **YOUR OWN** solutions. Do not blindly copy.

– Ask questions and come to office hrs if you get stuck. Don't let confusion snowball.

How to get an A/A+ in CSE251?

- ***Practice doing problems.*** Get comfortable with the math manipulations and associated physical meaning, and you will find exam problems to be easier
 - HW problems
 - Example problems worked in lecture and online class notes
 - Old exam problems
 - Office hours
- ***Review your prerequisites.***
 - Node analysis, Mesh Analysis, Circuit solving techniques! CSE250
- ***Come to class!!***
 - *HW & Participation are a significant part of your grade*
 - I will discuss topics to be emphasized on exams and give hints about how to approach the more difficult homework problems

How to ~~get an A/A+~~ pass in CSE251?

- ***Attendance + Assignments + Quiz + Lab:***

$$8\% + 12\% + 25\% + 15\% = 60\%$$

- Suppose, you attend all the classes. Get 83% in Assignments, 83% in Lab, 75% in Quiz. So, you will get:

$$8 + 10 + 21 + 11.25 = 50.25\% !!$$

- Try to do well in these continuous assessments and your road to passing CSE251 will be much easier.

- ***Come to class!!***

- *HW & Participation are a significant part of your grade*

- I will discuss topics to be emphasized on exams and give hints about how to approach the more difficult homework problems

How to approach CSE251?



Write down important information. **(See)**



Visualize - Draw - doodle -
interact **(Imagine)**



Think and solve. **(Act)**

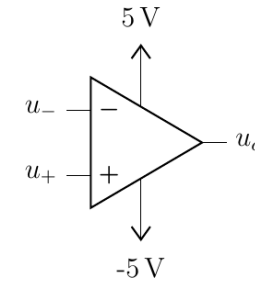


First Day Survey Spring 24

Course outline

Basically, study **four types** of devices.
(Application centric usage)

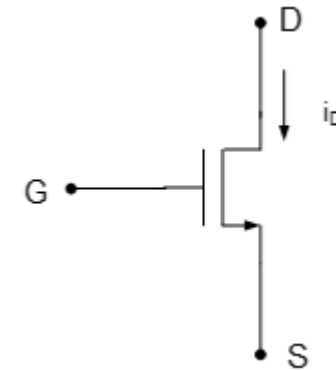
- OP-AMP



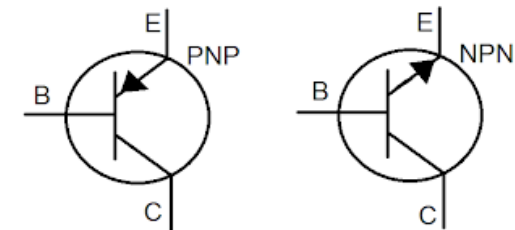
- Diodes



- MOSFET

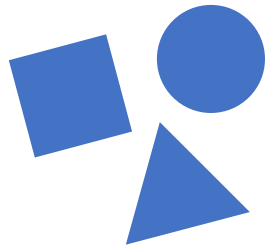


- BJT



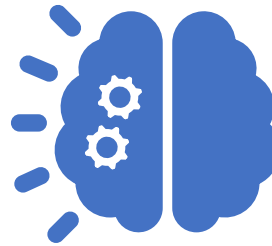
Application: **Amplification** and **Switching**

Course Outcome



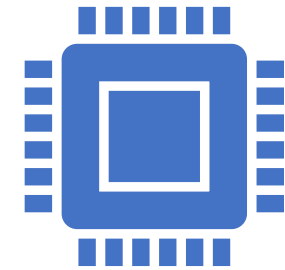
CO1

Understand and compare the characteristics and operation of electronic devices



CO2

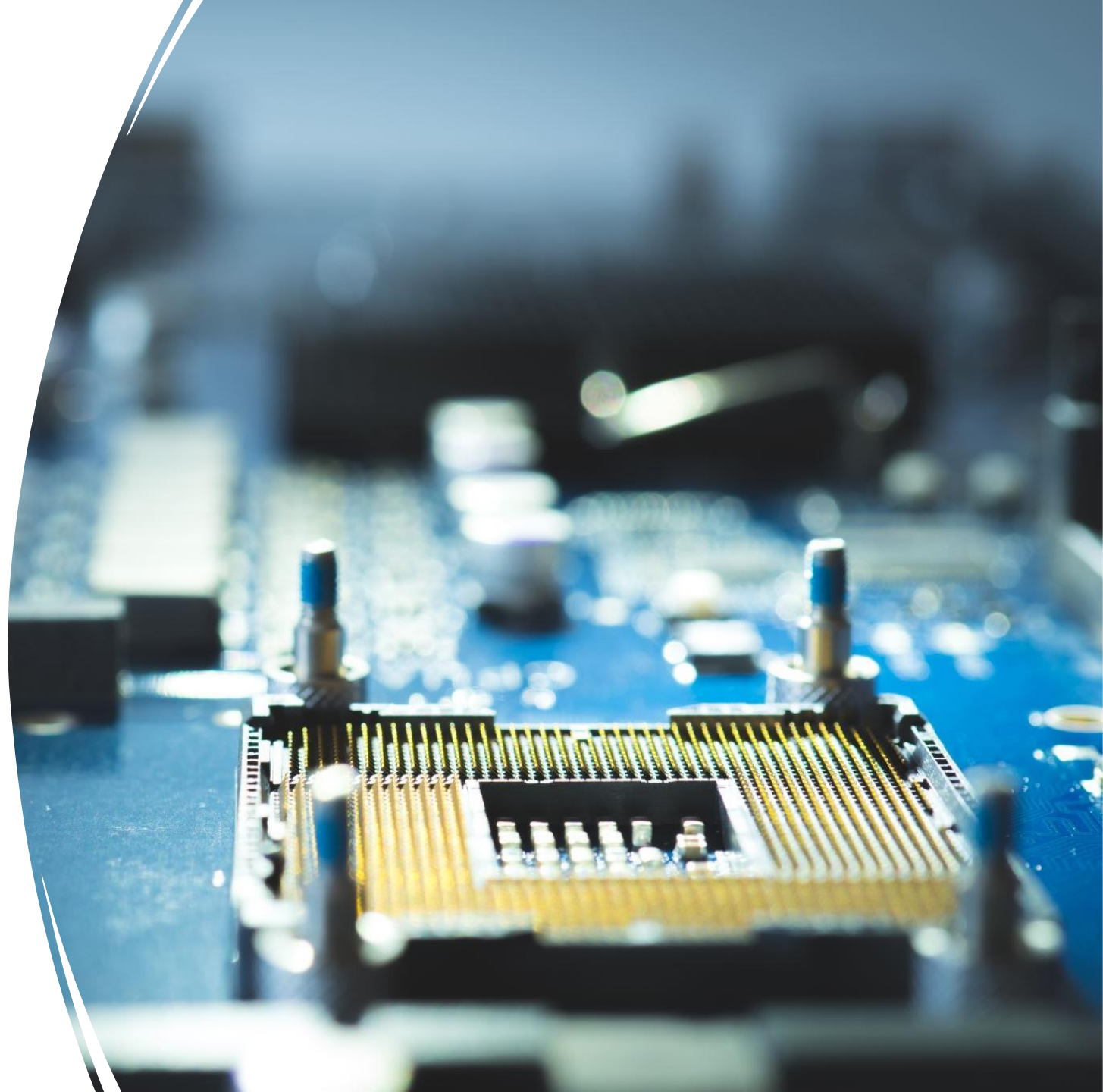
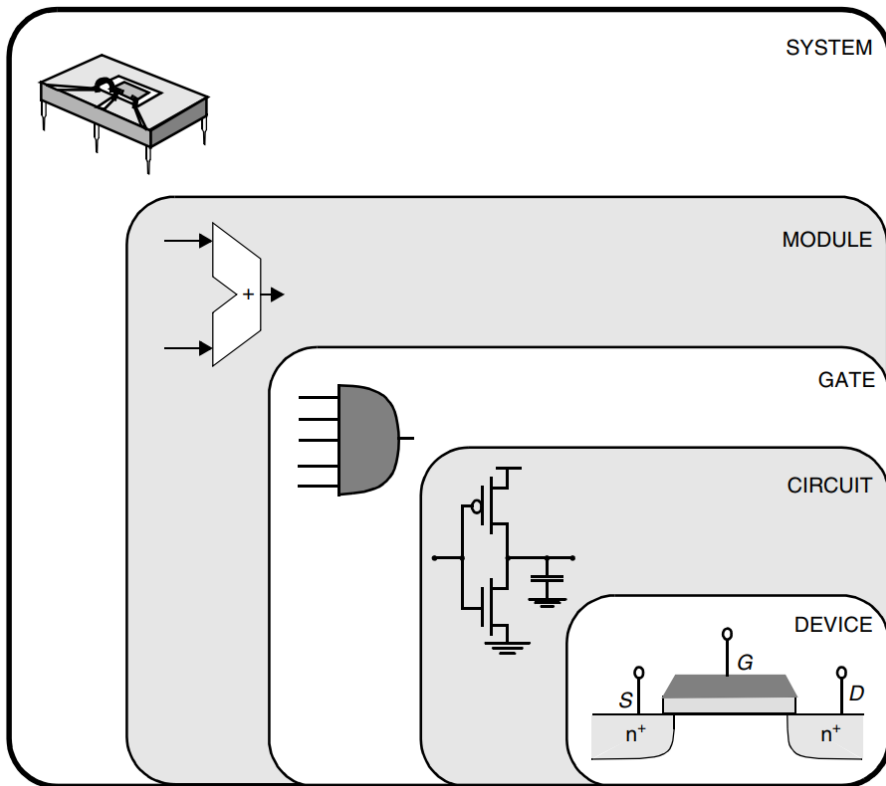
Analyze electronic circuits made from these devices



CO3

Design various electronic circuits for power-generation and analog signal-processing applications.

Abstraction Levels

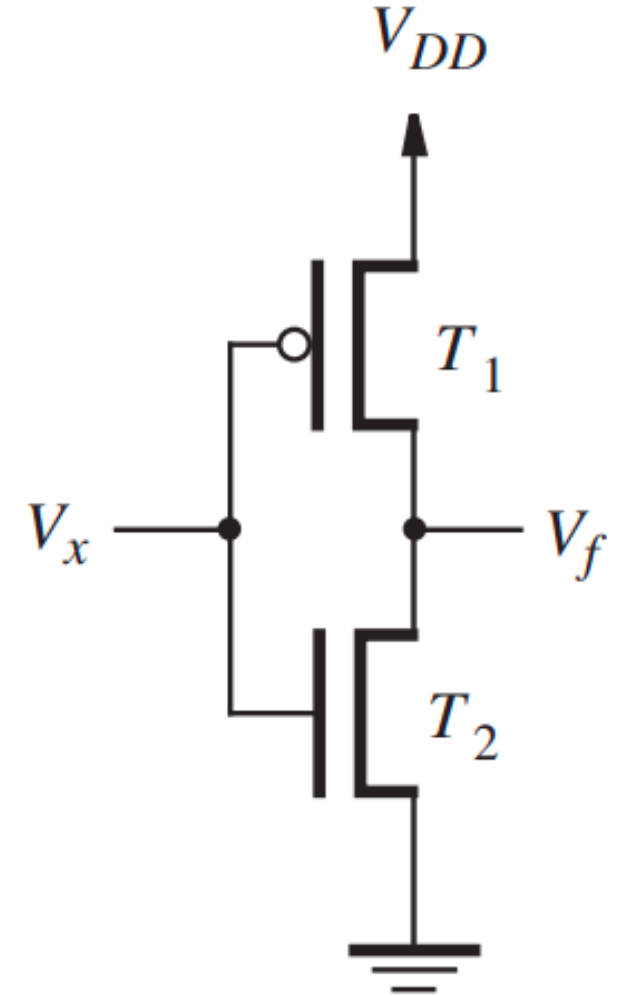


Logic gates → Electronic Devices

Electronic Devices:

1. Transistors (BJT/MOSFET/ JFET/**FinFET**)
2. Diodes

Amplification and **Switching**



MOSFET realization of a NOT gate.

Historical Perspective

- Has it always been like this?
- Eras of Computer evolution:

Mechanical gears
(1822 - [Difference Engine](#), Analytical Engine)

Electrical switches and
mechanical relays
(1944 - [Harvard Mark 1](#))

1. Gen 1: Mechanical to **Vacuum Tubes** (17th -1940s):

2. Gen 2: **Transistors** (BJT) (1950s): Short-lived

(1951 – 1959)
Switchover to **transistors** from
vacuum Tubes

3. Gen 3: **Integrated Circuitry** (1960s - Present)

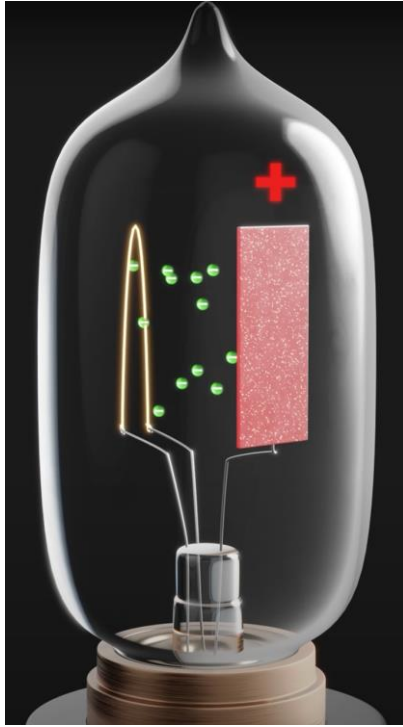
VLSI

[Microcomputers](#) -> Laptops, Smartphones

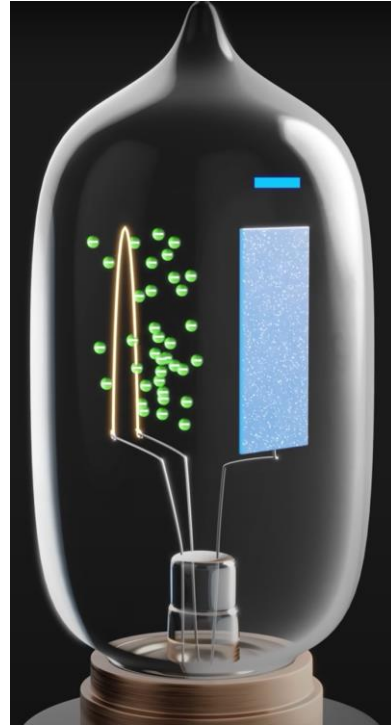
WWI: 1914 - 1918

WWII: 1939 - 1945

Vacuum Tubes (1946 - ENIAC)



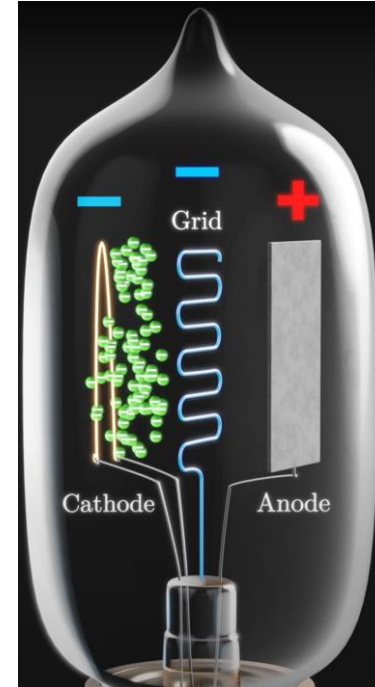
Electron Flow ON



Electron Flow OFF

Thermionic Diode

SWITCHING



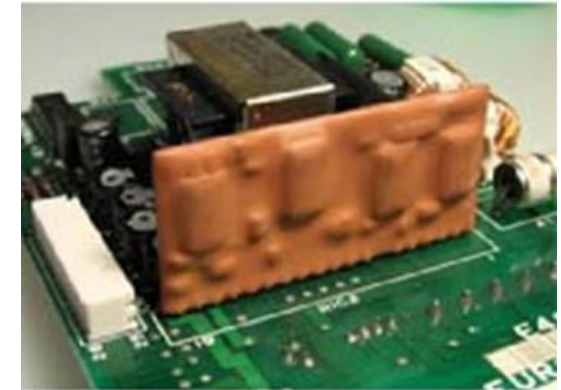
Small changes in **Grid** voltage translate to large voltages at the Anode

Thermionic Triode

AMPLIFICATION

Computer Systems /Processors with BJT (60s)

Computer System	Year
IBM System/360	1964
DEC PDP Series	1960 (PDP-1), 1965 (PDP-8), 1970 (PDP-11)
Control Data Corporation 6600	1964
IBM System/370	1970
Cray-1	1976



IBM System/360 hybrid BJT circuit

Computer Systems /Processors with MOSFET (70s - Present)



Intel® 4004

1971

10 μm



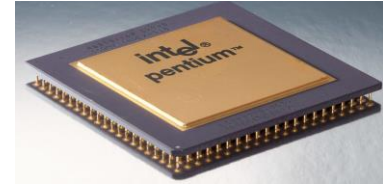
Intel® 8086

1978

3 μm



65 nm



0.8 μm

Intel® Pentium®

1993



22 nm

Intel® Core™

2010



7 nm

AMD Ryzen 5000

2021



Intel® Core™ 12th gen

2021

10 nm

Processor evolution

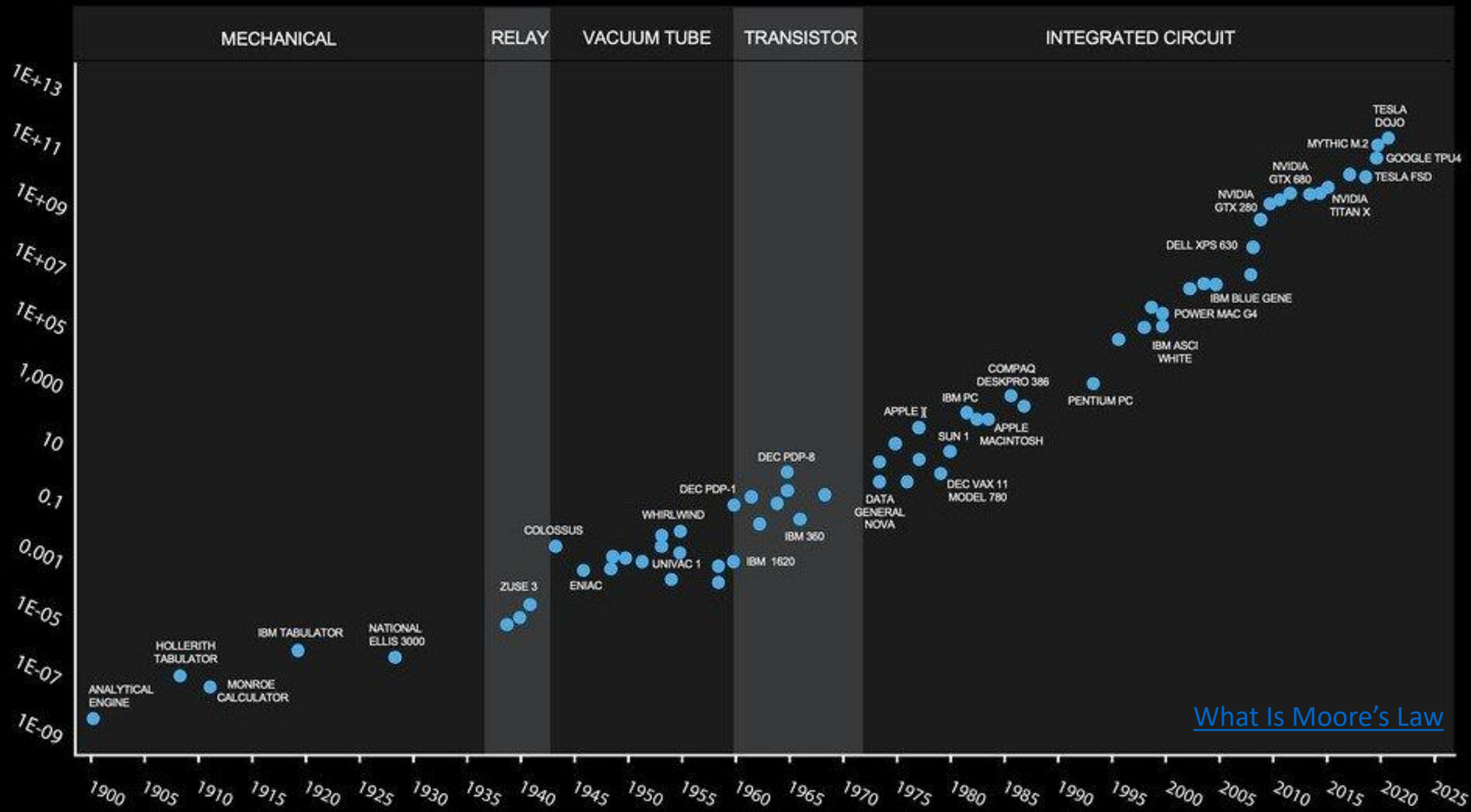
(Red: Feature size)

Moore's Law

Moore's Law

The number of transistors in a microchip doubles every two years

122 YEARS OF MOORE'S LAW



Current scenario and the future

- ~~Ongoing chip shortage!~~

- Two type of companies:

- 1. **Fabless** design companies: AMD
Apple etc.

- 2. **Foundries:** Intel
TSMC
Samsung
Global Foundries

Moore's Law is approaching an end.

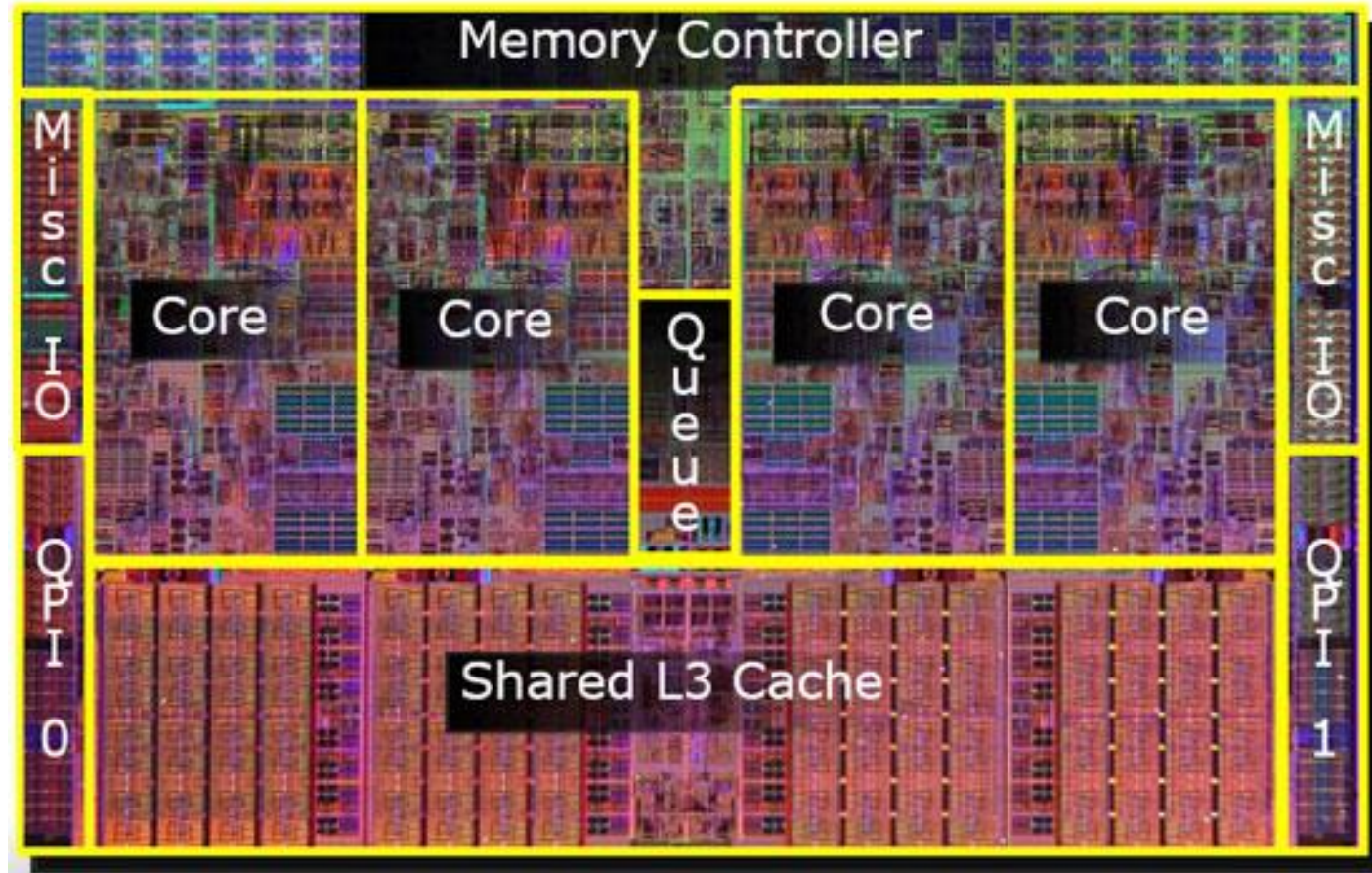
Possible alternatives for the future:

1. Spintronics

2. Photonics

3. Nano-electronics (Quantum)

An Integrated Circuit Layout of a Processor



List of resources used in this slide

- [History of Computers](#)
- [More about Vacuum Tubes – Veritasium YouTube](#)
- [More information about Semiconductor chip industry](#)
- [Moore's Law](#)

Thank You